

Exercise No.1 Solutions

Q.1 Explain the difference between safety and security in the context of machine learning systems. Give one concrete example of a safety concern and one security concern for an ML-based autonomous vehicle.

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Aspect	Safety	Security
Threat Source	Accidental failures, human error, or environmental hazards.	Intentional attacks and malicious actors.
Goal	Prevent unintentional harm or system failures.	Prevent unauthorized access or deliberate damage.
Focus	Robustness and fail-safe/fail-operational design.	Authentication, authorization, and encryption.

Examples for an ML-based Autonomous Vehicle (AV):

- ➔ **Safety Concern:** The perception model fails to detect a pedestrian because they are wearing a costume that wasn't in the training data (an accidental "out-of-distribution" failure).
- ➔ **Security Concern:** An attacker places specific "adversarial stickers" on a stop sign to make the ML model misclassify it as a speed limit sign

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Q.2 The EU AI Act optional The EU AI Act classifies AI systems into risk categories.

1. Name the main risk categories and briefly describe each.
2. Into which category would you place a perception system for
a) an autonomous vehicle b) a ChatBot?

Justify your answer.

3. Research: What obligations does the Act place on providers of systems in that category?

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1. Main Risk Categories

The Act classifies systems based on the potential harm they can cause:

- **Unacceptable Risk (Prohibited):** Systems that threaten safety or fundamental rights, such as social scoring or real-time biometric identification in public spaces.
- **High Risk (Highly Regulated):** Systems used in critical infrastructures (transport, medical, military, education) where failure could lead to significant harm.
- **Limited Risk (Requires Transparency):** Systems with a risk of manipulation, such as chatbots or LLMs, where users must be aware they are interacting with AI.
- **Minimal Risk (No Regulation):** Applications like spam filters or AI in video games.

2. Classification and Justification

- **AV Perception System: High Risk. Justification:** It is a safety component of a transport system. A failure here directly endangers human life and road safety.
- **ChatBot: Limited Risk. Justification:** The primary risk is the lack of transparency or potential for misinformation/manipulation, rather than direct physical harm.

3. Obligations for High-Risk Providers

Providers of High-Risk systems must comply with strict requirements, including:

- Establishing a risk management system throughout the entire lifecycle.
- Ensuring high-quality training and testing datasets to prevent bias.
- Detailed technical documentation and "logging" of system performance.
- Ensuring human oversight and providing a "right to explanation" for users.

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Q.3 Build the Right Model vs. Build the Model Right Two complementary perspectives on ML quality (or software quality in general) can be summarized as:

- “Build the right model” — correctness: does the model solve the intended problem?
- “Build the model right” — robustness: is the model built and deployed with adequate care?

For a pedestrian-detection model in an autonomous vehicle, describe one failure captured by each perspective.

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These two perspectives ensure overall system quality:

"Build the Right Model" (Correctness): Does the model solve the intended problem? This asks if our labels, objectives, and training data align with reality.

- **Failure Example:** A pedestrian detector that only recognizes people standing up. If it misses a person in a wheelchair because the training data lacked that "class," the model wasn't "built right" for the actual problem of road safety.

"Build the Model Right" (Robustness): Is the model built and deployed with adequate care? This focuses on the technical execution, such as preventing overfitting or ensuring it doesn't crash on noisy inputs.

- **Failure Example:** The model is highly accurate but extremely sensitive to sensor noise. A few "dead pixels" on the camera cause the model's confidence to drop to zero, leading to a missed detection.

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The Course Project: A Safety Case for a CARLA Perception Model

Throughout this course you will work towards a **safety case** for a concrete ML system. The system uses three separate binary classifiers trained on data collected in the **CARLA** autonomous driving simulator.

- **Input:** a single forward-facing camera image (shared across all three models).
- **Output:** three separate binary predictions, one from each classifier — *traffic light present, pedestrian present, vehicle present*.
- **Downstream use:** the models' outputs feed into the vehicle's autopilot decision logic.
- **Training data:** collected exclusively under *clear/sunny weather* and *daytime* conditions.

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A **safety case** is a structured argument, supported by evidence, that a system is acceptably safe in a given context (more on this in the second lecture). Each exercise sheet asks you to produce one piece of evidence. By the end of the course you will assemble these into a complete safety case.

Q.4 First Impressions Before looking at any data or model, think about the system described above.

1. Name three situations in which you would not trust this model's predictions. Explain why.
2. Which of the three output labels do you consider most safety-critical if missed? Justify
3. What additional information would you want before deploying this model in a real vehicle?

-> **1. Situations where the model might not be trusted**

Based on the fact that the model was trained **exclusively on sunny daytime data**:

- **Nighttime driving:** The features (colors, shadows) will be entirely different from the training set.
- **Heavy Rain or Fog:** Water on the lens or low visibility creates "out-of-distribution" visual noise.

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- **Unmapped/Rural roads:** The system is intended for "mapped intersections" and "urban environments"; it may fail on dirt roads or in forests.

2. Most safety-critical label

Pedestrian Present.

- **Justification:** While missing a traffic light or a vehicle is dangerous, pedestrians are the most "vulnerable road users." A vehicle has airbags and a metal frame for protection; a pedestrian has zero protection, making a missed detection highly likely to be fatal.

3. Additional information needed before deployment

- **Sensor Redundancy:** Since the system currently has **no radar or LiDAR**, I would want to know how the system handles direct sunlight blinding the camera (lens flare).
- **Human Fallback Data:** I would want to see data on operator reaction times during those **4-hour shifts** to see if "monitoring fatigue" is a major risk.

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Q.5 Course Topics and the Safety Case optional This seminar covers: testing and validation, explainability, uncertainty estimation, anomaly detection, adversarial machine learning, and alignment. For each topic, briefly explain how it could contribute evidence to the safety case for the CARLA model.

Topic	Contribution to the Safety Case
Testing/Validation	Provides quantitative evidence that the model meets performance requirements on held-out data.
Explainability	Evidence that the model is looking at the "right" features (e.g., the person, not just the sidewalk).
Uncertainty	Proves the model "knows when it doesn't know," allowing the system to hand back control to the human.
Anomaly Detection	Evidence that the system can identify "Out-of-Distribution" scenarios like snow or night and trigger a safe state.
Adversarial ML	Evidence that the model is robust against small, malicious perturbations in the input pixels.
Alignment	Ensures that the model's internal goals (e.g., minimizing loss) actually result in the safe behavior we want.